RAG: Baseline Retrieval vs. Cross-Encoder Re-Ranking

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Setup

Item	Value
Document	D&D Basic Rules (2018), Wizards of the Coast
Pages / chunks	180 pages → 1119 chunks
Chunking	774 chars mean, 150 overlap, page-bounded
Bi-encoder	`sentence-transformers/all-MiniLM-L6-v2`
Cross-encoder	`cross-encoder/ms-marco-MiniLM-L-6-v2`
Generator	`gemini-3-flash-preview`, temperature 0
Retrieve / context	top 10 candidates, top 3 to the LLM
Stress queries	15

Code and the executed notebook: `https://github.com/DeeptiKoranga/rag-baseline-vs-reranking`

1 Architecture Comparison

Measured quantities relevant to this section:

Quantity	Measured
Chunk vectors embedded once, reused every query	1119×384
Query embedding (one forward pass)	11.01 ms
Vector search over the whole corpus	0.07 ms
Cross-encoder, 10 pairs	57.30 ms
Cross-encoder, per (query, chunk) pair	5.7 ms

[Your analysis here.]

2 Qualitative and Rank Shift Analysis

2.1 How a failure is measured

For each query the chunk that actually answers it is pinned in advance by a regular expression matching a phrase unique to that chunk. Every claim below is therefore a rank, not an impression.

rank$_{100}$ is the answer-bearing chunk's position in a top-100 bi-encoder sweep (diagnostic only); *base* and *rerank* are its position among the 10 candidates before and after the cross-encoder. A dash means absent from that set. A query is **rescued** when the answer chunk was outside the top 3 for the baseline and inside it after re-ranking.

2.2 Full rank table

ID	Category	p.	rank$_{100}$	base	rerank	shift	Outcome
Q1	Keyword trap/distractor	46	6	6	7	−1	still missed
Q2	Keyword trap/distractor	54	2	2	1	+1	already in top 3
Q3	Keyword trap/distractor	63	1	1	2	−1	already in top 3
Q4	Exception/corner case	166	98	–	–	–	beyond recall
Q5	Exception/corner case	70	2	2	2	0	already in top 3
Q6	Exception/corner case	46	3	3	1	+2	already in top 3
Q7	Negation/constraint	77	2	2	1	+1	already in top 3
Q8	Negation/constraint	84	–	–	–	–	beyond recall
Q9	Negation/constraint	72	5	5	1	+4	**rescued**
Q10	Buried vital chunk	68	1	1	1	0	already in top 3
Q11	Buried vital chunk	67	6	6	1	+5	**rescued**
Q12	Buried vital chunk	67	2	2	2	0	already in top 3
Q13	Buried vital chunk	68	5	5	8	−3	still missed
Q14	Keyword trap/distractor	77	5	5	1	+4	**rescued**
Q15	Control	79	1	1	1	0	already in top 3

Summary	Count
Queries	15
Answer chunk in top 3 – baseline	8
Answer chunk in top 3 – re-ranked	11
Rescued by re-ranking	3
Never retrieved (re-ranking cannot help)	2
Retrieved but still outside top 3	2

2.3 Rank shifts, with what the baseline ranked instead

Q9 — Negation / constraint — Rank #5 → Rank #1

Query: *When am I prohibited from acting at all at the start of a battle?*

Why it is hard: Negative phrasing against a chapter full of 'you can' statements.

Baseline top 3 (cosine)

#	was	p.	chunk	cosine	text
1	–	72	509	0.419	nonplayer characters involved in combat, and each other pl
2	–	72	508	0.367	The clatter of a sword striking against a shield. The terr
3	–	162	1031	0.365	until the start of its next turn. Actions Greataxe. Melee

Re-ranked top 3 (cross-encoder)

#	was	p.	chunk	score	text
1	#5	72	511	-7.001	If you're surprised, you can't move or take an action on y ← **ANSWER**
2	#3	162	1031	-8.203	until the start of its next turn. Actions Greataxe. Melee
3	#9	73	517	-9.533	The DM might require you to use an action for any of these

Q11 — Buried vital chunk — Rank #6 → Rank #1

Query: *Does scrambling up a wall or wading through a swamp slow me down?*

Why it is hard: The movement-cost rule is phrased in feet, not in 'slow down'.

Baseline top 3 (cosine)

#	was	p.	chunk	cosine	text
1	–	66	459	0.395	Movement Swimming across a rushing river, sneaking down a
2	–	73	520	0.384	If you have more than one speed, such as your walking spee
3	–	73	522	0.377	A. You can drop prone without using any of your speed. Sta

Re-ranked top 3 (cross-encoder)

#	was	p.	chunk	score	text
1	#6	67	468	-8.171	Difficult Terrain The travel speeds given in the Travel Pa ← **ANSWER**
2	#1	66	459	-8.270	Movement Swimming across a rushing river, sneaking down a
3	#5	73	521	-8.879	treacherous staircases—the setting of a typical fight cont

Q14 — Keyword trap / distractor — Rank #5 → Rank #1

Query: *Can I attack with a weapon in each hand on the same turn?*

Why it is hard: The Fighter class has a feature *called* 'Two-Weapon Fighting', which is not the rule – the rule, with its 'light' restriction, is 50 pages later in the combat chapter.

Baseline top 3 (cosine)

#	was	p.	chunk	cosine	text
1	–	27	193	0.403	Two-Weapon Fighting When you engage in two-weapon fighting
2	–	77	550	0.389	unless that modifier is negative. If either weapon has the
3	–	27	197	0.381	or (b) two handaxes • (a) a dungeoneer's pack or (b) an ex

Re-ranked top 3 (cross-encoder)

#	was	p.	chunk	score	text
1	#5	77	549	2.119	modifier—the same modifier used for the attack roll—to the ← **ANSWER**
2	#1	27	193	1.577	Two-Weapon Fighting When you engage in two-weapon fighting
3	#7	73	516	1.445	Other Activity on Your Turn Your turn can include a variet

2.4 Cases where re-ranking did not help

- **Q4** (Exception / corner case) — *beyond recall.* The answer chunk on p. 166 sits at rank 98 of a top-100 sweep, so it never enters the 10 candidates and no re-ranker can reach it. Query: *If a fight comes at us in several waves, can we rest in between to heal up?*
- **Q8** (Negation / constraint) — *beyond recall.* The answer chunk on p. 84 does not appear in a top-100 sweep at all, so it never enters the 10 candidates and no re-ranker can reach it. Query: *What will make me lose a spell I am trying to keep going?*
- **Q1** (Keyword trap / distractor) — *still missed.* Retrieved at #6, re-ranked to #7, still outside the top 3. Query: *Does wearing heavy plate make it harder to sneak past guards unnoticed?*
- **Q13** (Buried vital chunk) — *still missed.* Retrieved at #5, re-ranked to #8, still outside the top 3. Query: *How much damage do you take from a long drop?*

2.5 Answer 1 vs. Answer 2

Q14 — Can I attack with a weapon in each hand on the same turn?

Answer 1 (baseline) — context pages [27, 77, 27], chunks [193, 550, 197]:

> *Based on the provided context, you can engage in two-weapon fighting. When doing so, you can add your ability modifier to the damage of the second attack, unless that modifier is negative (p. 27, 77). If either weapon has the thrown property, you may throw it instead of making a melee attack (p. 77). Additionally, if you are at least 2nd level, you can use Action Surge to take one additional action on your turn, which increases to two uses at 17th level (p. 27). You must finish a short or long rest before using Action Surge again (p. 27).*

Answer 2 (re-ranked) — context pages [77, 27, 73], chunks [549, 193, 516]:

> *Yes, when you take the Attack action with a light melee weapon in one hand, you can use a bonus action to attack with a different light melee weapon in the other hand (p. 77). If either weapon has the thrown property, you may throw it instead of making a melee attack (p. 77). Restrictions and penalties include: * Both weapons must be light melee weapons (p. 77). * You do not add your ability modifier to the damage of the bonus attack unless the modifier is negative (p. 77). However, certain class features allow you to add the modifier to the second attack's damage (p. 27).*

Q9 — When am I prohibited from acting at all at the start of a battle?

Answer 1 (baseline) — context pages [72, 72, 162], chunks [509, 508, 1031]:

> *NOT IN CONTEXT.*

Answer 2 (re-ranked) — context pages [72, 162, 73], chunks [511, 1031, 517]:

> *You are prohibited from moving or taking an action on your first turn of combat if you are surprised (p. 72). Additionally, you are restricted from taking any reactions until that first turn ends (p. 72). While a reaction is an instant response to a trigger that can occur on your turn or someone else's, once you use a reaction, you are prohibited from taking another one until the start of your next turn (p. 73).*

Q11 — Does scrambling up a wall or wading through a swamp slow me down?

Answer 1 (baseline) — context pages [66, 73, 73], chunks [459, 520, 522]:

> *Yes. According to the rules for difficult terrain, every foot of movement costs 1 extra foot (p. 73). The context identifies boulder-strewn caverns, briar-choked forests, and treacherous staircases as examples of settings containing difficult terrain (p. 73). Additionally, scaling a treacherous mountain slope or swimming across a rushing river are noted as key types of movement in adventures (p. 66). While the DM may summarize movement between encounters without calculating exact distances, it is sometimes important to determine how long it takes to travel between spots in days, hours, or minutes (p. 66).*

Answer 2 (re-ranked) — context pages [67, 66, 73], chunks [468, 459, 521]:

Yes. Climbing and swimming (such as wading through a swamp) are special types of movement where each foot costs 1 extra foot of speed (p. 67). If the terrain is considered difficult terrain—such as deep swamps, steep mountains, or rubble—you move at half speed, meaning 1 foot of movement costs 2 feet of speed (p. 67). When climbing or swimming in difficult terrain, each foot costs 2 extra feet (p. 67). Additionally, the travel speeds for simple terrain assume you can cover only half the normal distance in a minute, hour, or day when in difficult terrain (p. 67). This rule applies even if multiple factors in a space count as difficult terrain (p. 73).

[Your analysis here.]

3 Latency Profiling

15 queries × 5 repeats = 75 measurements per pipeline, after one untimed warm-up pass. The LLM call is excluded from the pipeline comparison: it is identical work in both pipelines and its variance is far larger than the retrieval times under test. It is reported separately below.

Stage	**mean** (ms)	**median** (ms)	**p95** (ms)	**min** (ms)	**max** (ms)
Query embedding (1 pass)	11.01	11.18	12.21	6.84	13.71
Vector search (whole corpus)	0.07	0.06	0.07	0.05	0.37
Cross-encoder (10 pairs)	57.3	48.34	110.96	37.45	114.11
Pipeline 1 retrieval	**11.08**	11.24	12.32	6.9	13.77
Pipeline 2 retrieval	**68.38**	59.99	121.62	44.87	125.68

Pipeline 2 retrieval costs **6.2**× Pipeline 1 (+**57** ms absolute). Measured separately, the LLM call averages **34674** ms over 6 calls, so against generation the re-ranker adds **0.2**% to end-to-end latency — the generator dominates by roughly 605×.

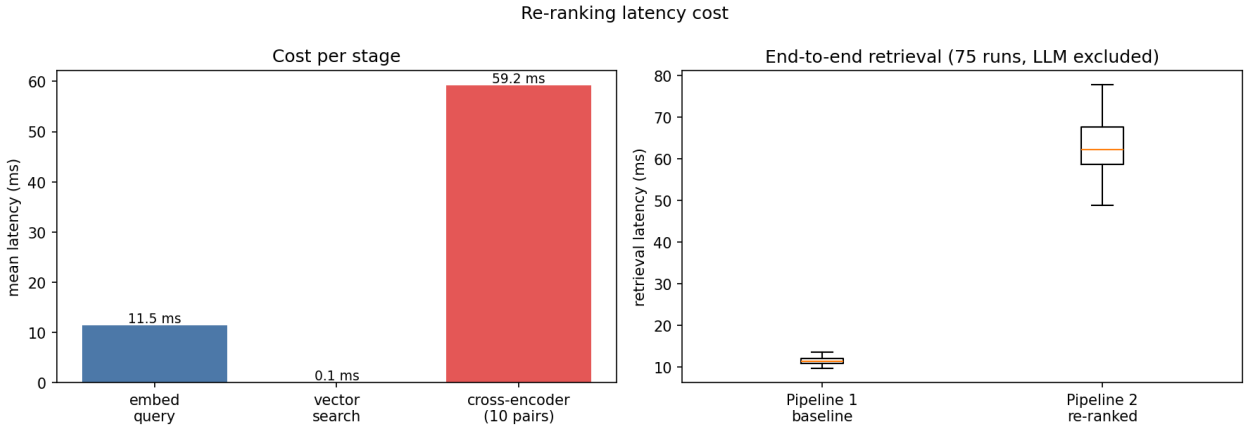


Figure 1: Per-stage cost (left) and end-to-end retrieval latency (right).

[Your analysis here.]

4 Scaling: Why Not Cross-Encode Everything?

The cross-encoder costs 5.7 ms per (query, chunk) pair and is linear in the number of pairs, because a score exists only for a specific pair and nothing can be precomputed between queries.

Corpus	**Chunks**	**Cross-encode all** (s/query)	**Two-stage** (s/query)	**Ratio**
This rulebook	1,119	6.4	0.068	94×
10k chunks	10,000	57.3	0.068	838×
1M chunks	1,000,000	5,730.0	0.068	83,796×

[Your analysis here.]